



SOMAIYA
VIDYAVIHAR UNIVERSITY

K J Somaiya School of Engineering
(formerly K J Somaiya College of Engineering)



Unit 2.2

Pulse Code Modulation

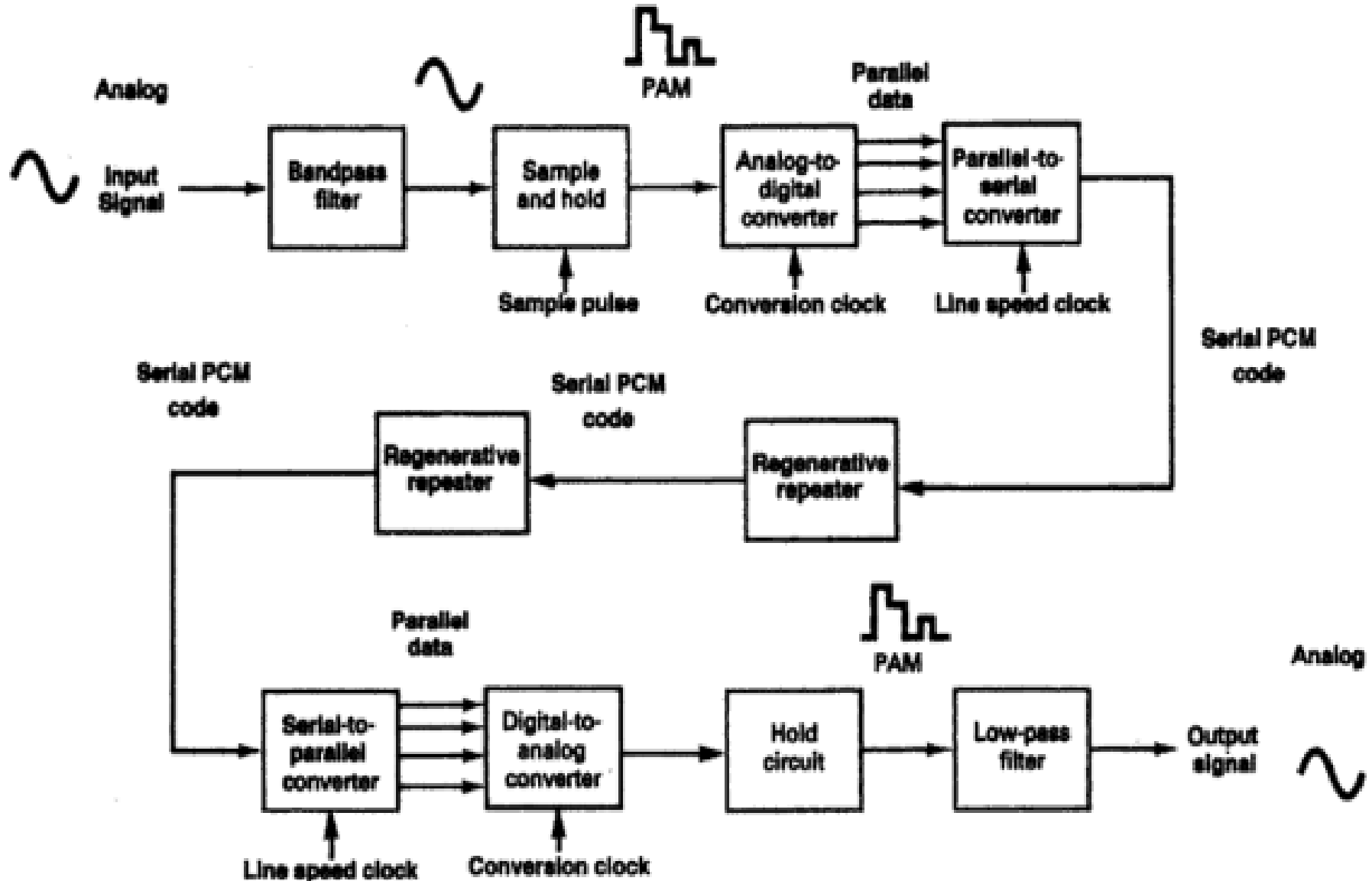
Introduction

- Pulse Code Modulation (PCM) is a method for digitally representing analog signals
- PCM involves sampling the analog signal,
- quantizing the samples to discrete levels,
- And then encoding these levels into a binary sequence (a series of 0s and 1s)

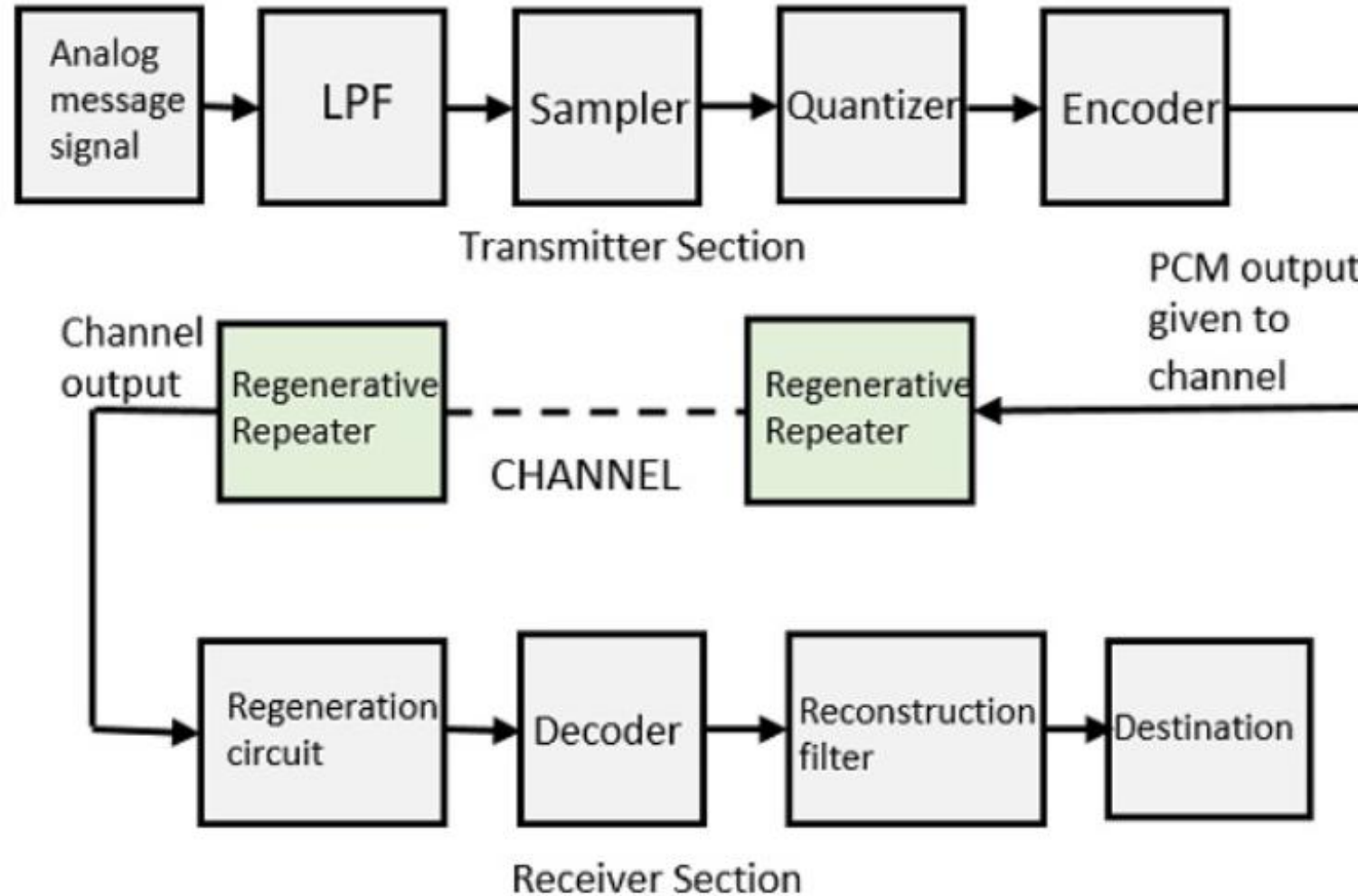
Introduction

- **Sampling:** The analog signal is measured at regular intervals, creating a series of instantaneous values
- **Quantization:** These sampled values are then rounded off to the nearest level from a predefined set of discrete levels
- **Encoding:** The quantized values are represented using a binary code (sequence of 0s and 1s)

PCM Transmission System



PCM Transmission System



Low Pass Filter

- This eliminates the high frequency components present in the input analog signal,
- Which is greater than the highest frequency of the message signal,
- To avoid aliasing of the message signal

Sampler

- Helps to collect the sample data at instantaneous values of the message signal,
- So as to reconstruct the original signal
- The sampling rate must be greater than twice the highest frequency component f_m of the message signal (as per sampling theorem)

Quantizer

- It is a process of reducing the excessive bits and confining the data
- The sampled output when given to Quantizer,
- It reduces the redundant bits and compresses the value

Encoder

- The digitization of analog signal is done by the encoder
- Designates each quantized level by a binary code
- Sample-and-hold circuit is used
- Encoding minimizes the bandwidth use

Regenerative Repeater

- Increases the signal strength
- It compensate the signal loss and reconstruct the signal,
- And also to increase its strength

Decoder

- Decodes the pulse coded waveform to reproduce the original signal
- This circuit acts as the demodulator

Reconstruction Filter

- It is a low-pass filter
- It gets back the original signal

Applications of PCM

- **Digital audio:** Used for digital audio in computer, CDs,
- **Telephony:** Digital telephone systems to transmit voice signals
- **Robustness:** Improved resistance to noise and interference compared to analog transmission
- **Digital Processing:** the binary representation of the signal allows for digital signal processing
- **Standardization:** Adopted as standard for digital encoding of waveforms